

Education

- **B.Sc in Computer Science and Engineering**, Islamic University of Technology, Dhaka **2023 – Present**
 - CGPA: 3.91 / 4.00
- **Higher Secondary Certificate, Science**, Notre Dame College, Dhaka **2020 – 2022**
 - GPA: 5.00 / 5.00
- **Secondary School Certificate, Science**, St. Joseph High School, Dhaka **2018 – 2020**
 - GPA: 5.00 / 5.00

Projects

IUT Cafeteria Crisis

2026

- Migrated a monolith into 5 containerized microservices deployed via Docker and Kubernetes.
- Prevented database deadlocks and over-selling using PostgreSQL optimistic locking and Redis caching.
- Engineered a resilient order-processing queue using RabbitMQ and WebSockets for real-time updates.
- Implemented UUID idempotency keys to prevent double-charging and secure transaction reliability.
- Integrated Prometheus for metrics, Jaeger for distributed tracing, and custom chaos testing endpoints.
- Technologies used: TypeScript, Node.js, React, PostgreSQL, Redis, RabbitMQ, Docker, Kubernetes, GitHub Actions, Prometheus, Jaeger.

The Existential Choreography Planner

2026

- Developed a full-stack application utilizing React Flow to transform daily chores into interactive, drag-and-drop dependency flowcharts.
- Created a custom backend engine with Node.js and MongoDB to track user energy regeneration, process complex task dependencies, and award gamified badges.
- Integrated secure JWT and Google OAuth authentication, and engineered robust functionality for JSON-based workflow import and export.
- Technologies used: React, Tailwind CSS, React Flow, TypeScript, MongoDB, Docker, OAuth

Enterprise Gym Management System

2026

- Built a comprehensive gym management SPA using a layered Spring Boot backend and React/TypeScript.
- Implemented REST APIs to manage member subscriptions, class scheduling, billing, and equipment inventory.
- Integrated stateless JWT authentication and orchestrated seamless deployment using Docker Compose.
- Technologies used: Java, Spring Boot, React, TypeScript, PostgreSQL, Docker

Technical Skills

- **Languages:** Java, Python, C++, C#, HTML, CSS, JavaScript
- **Frameworks/Libraries:** React, Spring Boot, Express, OAuth, SFML, SDL
- **Databases :** MongoDB, PostgreSQL
- **Tools :** Redis, RabbitMQ, Docker, Kubernetes, Git, Prometheus
- **Concepts :** OOP, DSA, DBMS

Soft Skills

Leadership • Event Management • Collaboration • Problem-Solving • Adaptability • Fluency in English